



Roll Number

JIS UNIVERSITY
2025

END TERM EXAMINATIONS, AY 2024 - 2025(EVEN)
YMT4001 - PROBABILITY AND STATISTICS

Time: 2 Hrs

Maximum Marks : 50

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any Ten (10x1=10 Marks)**

Find:

(1) CO5 BL1 PO1

Two events A and B are said to be independent if:

(A) $P(A \cap B) = P(A) + P(B)$

(B) $P(A|B) = P(A)$

(C) $P(A \cap B) = P(A) - P(B)$

(D) $P(A|B) = 1 - P(A)$

2. Solve:

(1) CO3 BL2 PO4

If the average number of calls at a call center per minute follows a Poisson distribution with mean 3, what is the probability of receiving exactly 2 calls in a minute?

(A) $\frac{3^2 e^{-3}}{2!}$

(B) $\frac{3^2}{2!}$

(C) $3e^{-3}$

(D) None of the above

3. Solve:

(1) CO1 BL1 PO1

A probability mass function (PMF) must satisfy which of the following properties?

- (A) $P(X = x) \geq 0$ for all x
- (B) $\sum P(X = x) = 1$
- ☒ (C) Both A and B
- (D) None of the above

4. Solve:

(1) CO1 BL1 PO1

If $\text{Var}(X) = 4$, what is $\text{Var}(3X)$?

- ☒ (A) 12
- (B) 9
- (C) 36
- (D) None of the above

5. Find:

(1) CO2 BL2 PO1

A discrete random variable X takes the values x_i with probabilities $\frac{K}{5^i}$.
The value of K is

- ☒ (A) $\frac{4}{5}$
- (B) $\frac{5}{4}$
- (C) $\frac{45}{54}$
- (D) None of the above

6. Find:

(1) CO1 BL3 PO1

If X is a random variable and a is a constant, what is $E(aX)$?

- (A) $E(X)$
- ☒ (B) $aE(X)$
- (C) $E(X) + a$
- (D) None of the above.

7. Find:

(1) CO1 BL1 PO1

The variance of a Poisson-distributed random variable $X \sim \text{Poisson}(\lambda)$ is:

- (A) λ
 (B) λ^2
 (C) $e^{-\lambda}$
 (D) None of the above.

Solve:

(1) CO2 BL2 PO1

If a fair coin is flipped 10 times, what is the probability of getting exactly 5 heads?

- (A) $\binom{10}{5}(0.5)^5(0.5)^5$
 (B) $\binom{10}{5}(0.5)^{10}$
 (C) $(0.5)^5$
 (D) None of the above.

Find:

(1) CO4 BL1 PO1

If $E(X) = 10$ and $E(Y) = 5$, then $E(2X + 3Y)$ is:

- (A) 35
 (B) 40
 (C) 30
 (D) None of the above

8. If the events A and B are independent, which one is true:

(1) CO1 BL1 PO1

- (A) complement of A and B are independent
 (B) A and complement of B are independent
 (C) complement of A and complement of B are independent
 (D) All of the above

9.

(1) CO1 BL1 PO1

Solve

If $P(A|B) = 0.5$ and $P(B) = 0.4$, what is $P(A \cap B)$?

- (A) 0.2
 (B) 0.4
 (C) 0.5
 (D) 0.1

12. Solve:

(1) CO2 BL3 PO4

If X is a discrete random variable with PMF given by $P(X = x) = \frac{1}{k}$ for $x = 1, 2, 3, 4, 5$, what is the value of k ?

- (A) 5
 (B) 10
 (C) 15
 (D) None of the above

Part B

Answer any Two (2x5=10 Marks)

13. Find the value of k such that $f(x) = kx$ where $1 \leq x \leq n$ is a probability mass function of a discrete random variable X . Find the mean and variance.

(5) CO4 BL3 PO

14. (a) (2 marks)

(5) CO2 BL4 PO

Suppose X is a random variable which follows Poisson distribution. If $P(X=0) = P(X=2)$, find $P(X=1)$.

(b) (3 marks)

A fair coin is tossed until a head or four tails occur. If X denotes the number of tosses, find mean and variance of X .

15. Consider the following:

(5) CO4 BL2 PO

Let X be a random variable which follows normal distribution with mean 3 and variance 4. Find the value of c such that

$$P(X > c) = 2P(X \leq c)$$

Given that

$$\int_{-\infty}^{-0.43} \phi(z) dz = \frac{1}{3}$$

where $\phi(z)$ is the standard normal density function.

16. Jennifer throws a dice. If she gets a prime number, she tosses a coin 3 times and counts the number of tails. If she gets any other number, she tosses a coin only once. Now Jennifer says she got exactly one tail, find the probability she got a prime number.

(5) C02 BL2 P04

Part C

Answer any Three (3x10=30 Marks)

7. (a) (7 marks) (5+2)

(10) C03 BL1 P04

Let M denote the mean and V denote the variance of a binomial distribution. If sum of M and V is 50 and their product is 600, determine the parameters n, p, q of binomial distribution. Hence calculate $P(X=0)$.

- (b) (3 marks)

In a game a man wins two rupees for a six and loses a rupee for any other number when a fair dice is thrown. The man decided to throw a dice thrice but to quit as and when he gets a six. Find the expected value of the amount he wins/loses.

8. (a) (6 marks)

(10) C04 BL2 P04

The heights of adult males in a city follow a normal distribution with mean 175 cm and standard deviation 8 cm. Find the probability that a randomly selected male is taller than 185 cm. Also, find the probability that a randomly selected male is shorter than 185 cm.

Given $P(Z < 1.25) = 0.8944$

- (b) (4 marks)

Calculate Spearman's Rank Correlation for the following data set consisting of 6 values:

X

Y

34	80
67	92
24	67
40	78
54	38
89	91

(10) CO5 BL2 PO2

19. (a) (3 marks)

Determine the value of k such that $f(x)$ defined by

$$f(x) = \begin{cases} ke^{-b(x-a)}, & a < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

is a probability density function (PDF).

(7 marks)

Also calculate the variance.

(10) CO5 BL1 PO2

20. (a) (4 marks)

Consider the following data:

$$\sum x = 140, \quad \sum y = 150, \quad \sum (x-10)^2 = 180, \quad \sum (y-15)^2 = 215,$$

$$\sum (x-10)(y-15) = 60, \quad n = 10$$

calculate the correlation coefficient.

(b) (6 marks)

If $k = \frac{1}{2}$ and $f(x)$ is a Probability Density Function, find the distribution function.

$$f(x) = \begin{cases} kx, & 0 < x < 1 \\ k, & 1 \leq x \leq 2 \\ -kx + 3k, & 2 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$$

Also, find $P(0.5 < X < 1)$.

19/05/2025, 12:14 PM
(a) (6 marks)

Check if the lines $3X+9Y=17$ and $31X+12Y=7$ are regression lines. If yes, calculate the value of correlation coefficient wherever applicable.

(b) (4 marks)

A biased dice in which the probability of getting an prime number is thrice that of any other number is thrown 250 times. If getting a prime number is considered a success, find the mean and variance.

CO Definitions:

- To explain and demonstrate the distinctive characteristics of probability distribution.
- To analyze the probability of real-world uncertain phenomena by identifying probability distribution that fits the phenomena.
- To explain and demonstrate the distinctive characteristics of statistics.
- To apply and analyze the uses and limitations of statistical analysis
- Apply probability and statistical concepts to solve problems in artificial intelligence, data science, and other relevant fields